

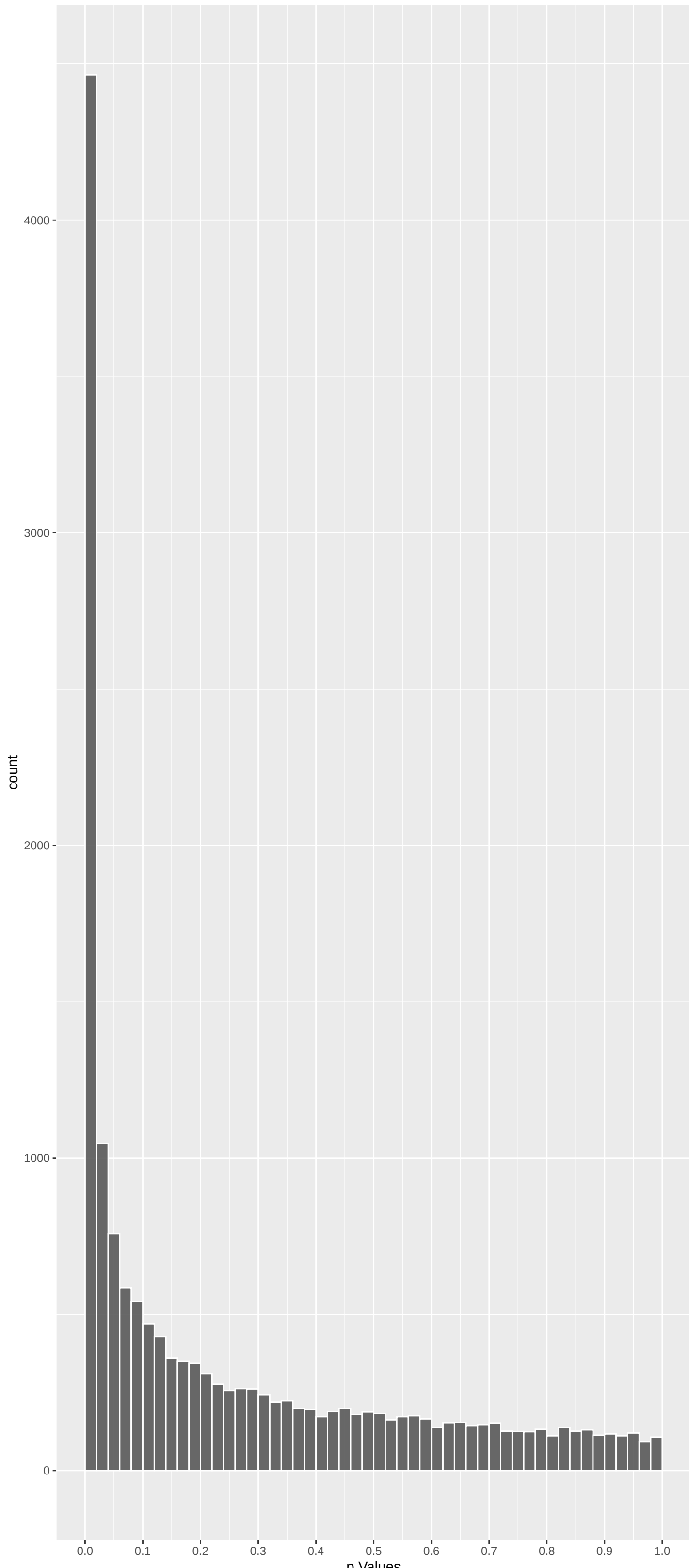
Top genes by P–value non–permuted

Gene	Rho	N	P	p.adj	qValueNoperm
SPG11	0.09921964	627	3.256e-14	5.252e-10	2.040e-06
TRPM5	0.11837287	511	7.777e-14	6.273e-10	2.040e-06
EP400	0.10360299	572	1.426e-13	7.669e-10	2.040e-06
ZFAND4	0.11180013	517	2.972e-13	1.199e-09	2.391e-06
TRIM15	0.13087578	423	1.016e-12	3.279e-09	3.432e-06
PLEC	0.11871026	461	1.388e-12	3.731e-09	3.432e-06
ADGRF1	0.08909133	604	2.160e-12	3.871e-09	3.432e-06
ECI2	0.10166850	530	2.095e-12	3.871e-09	3.432e-06
TMEM131L	0.09193039	590	1.746e-12	3.871e-09	3.432e-06
SLC14A2	0.09200406	573	3.735e-12	6.026e-09	4.421e-06
PJA2	0.10292213	506	5.172e-12	7.584e-09	4.421e-06
CLCNKA	0.10289648	501	6.732e-12	7.757e-09	4.421e-06
AKAP9	0.08725110	592	6.349e-12	7.757e-09	4.421e-06
DNAH14	0.10140921	509	6.508e-12	7.757e-09	4.421e-06
MTBP	0.08607087	588	1.069e-11	1.150e-08	6.115e-06
MMEL1	0.09778120	502	2.303e-11	2.322e-08	1.060e-05
AGGF1	0.09764501	501	2.502e-11	2.374e-08	1.060e-05
CLCNKB	0.10236078	475	2.912e-11	2.524e-08	1.060e-05
SPEGNB	0.21077320	231	2.973e-11	2.524e-08	1.060e-05
RIMBP2	0.08794335	541	4.873e-11	3.931e-08	1.568e-05
CLDN16	0.12205728	382	7.977e-11	6.057e-08	2.101e-05
CFAP20DC	0.08305575	559	8.636e-11	6.057e-08	2.101e-05
FGD5	0.08966837	518	8.578e-11	6.057e-08	2.101e-05
TACR2	0.10287762	446	1.145e-10	7.698e-08	2.559e-05
PPP4R1	0.08781879	516	1.511e-10	9.748e-08	3.111e-05

Top genes by Q–Value non–permuted

Gene	Rho	N	P	p.adj	qValueNoperm
SPG11	0.09921964	627	3.256e-14	5.252e-10	2.040e-06
EP400	0.10360299	572	1.426e-13	7.669e-10	2.040e-06
TRPM5	0.11837287	511	7.777e-14	6.273e-10	2.040e-06
ZFAND4	0.11180013	517	2.972e-13	1.199e-09	2.391e-06
PLEC	0.11871026	461	1.388e-12	3.731e-09	3.432e-06
ADGRF1	0.08909133	604	2.160e-12	3.871e-09	3.432e-06
TRIM15	0.13087578	423	1.016e-12	3.279e-09	3.432e-06
ECI2	0.10166850	530	2.095e-12	3.871e-09	3.432e-06
TMEM131L	0.09193039	590	1.746e-12	3.871e-09	3.432e-06
CLCNKA	0.10289648	501	6.732e-12	7.757e-09	4.421e-06
AKAP9	0.08725110	592	6.349e-12	7.757e-09	4.421e-06
PJA2	0.10292213	506	5.172e-12	7.584e-09	4.421e-06
DNAH14	0.10140921	509	6.508e-12	7.757e-09	4.421e-06
SLC14A2	0.09200406	573	3.735e-12	6.026e-09	4.421e-06
MTBP	0.08607087	588	1.069e-11	1.150e-08	6.115e-06
AGGF1	0.09764501	501	2.502e-11	2.374e-08	1.060e-05
CLCNKB	0.10236078	475	2.912e-11	2.524e-08	1.060e-05
MMEL1	0.09778120	502	2.303e-11	2.322e-08	1.060e-05
SPEGNB	0.21077320	231	2.973e-11	2.524e-08	1.060e-05
RIMBP2	0.08794335	541	4.873e-11	3.931e-08	1.568e-05
CLDN16	0.12205728	382	7.977e-11	6.057e-08	2.101e-05
CFAP20DC	0.08305575	559	8.636e-11	6.057e-08	2.101e-05
FGD5	0.08966837	518	8.578e-11	6.057e-08	2.101e-05
TACR2	0.10287762	446	1.145e-10	7.698e-08	2.559e-05
PPP4R1	0.08781879	516	1.511e-10	9.748e-08	3.111e-05

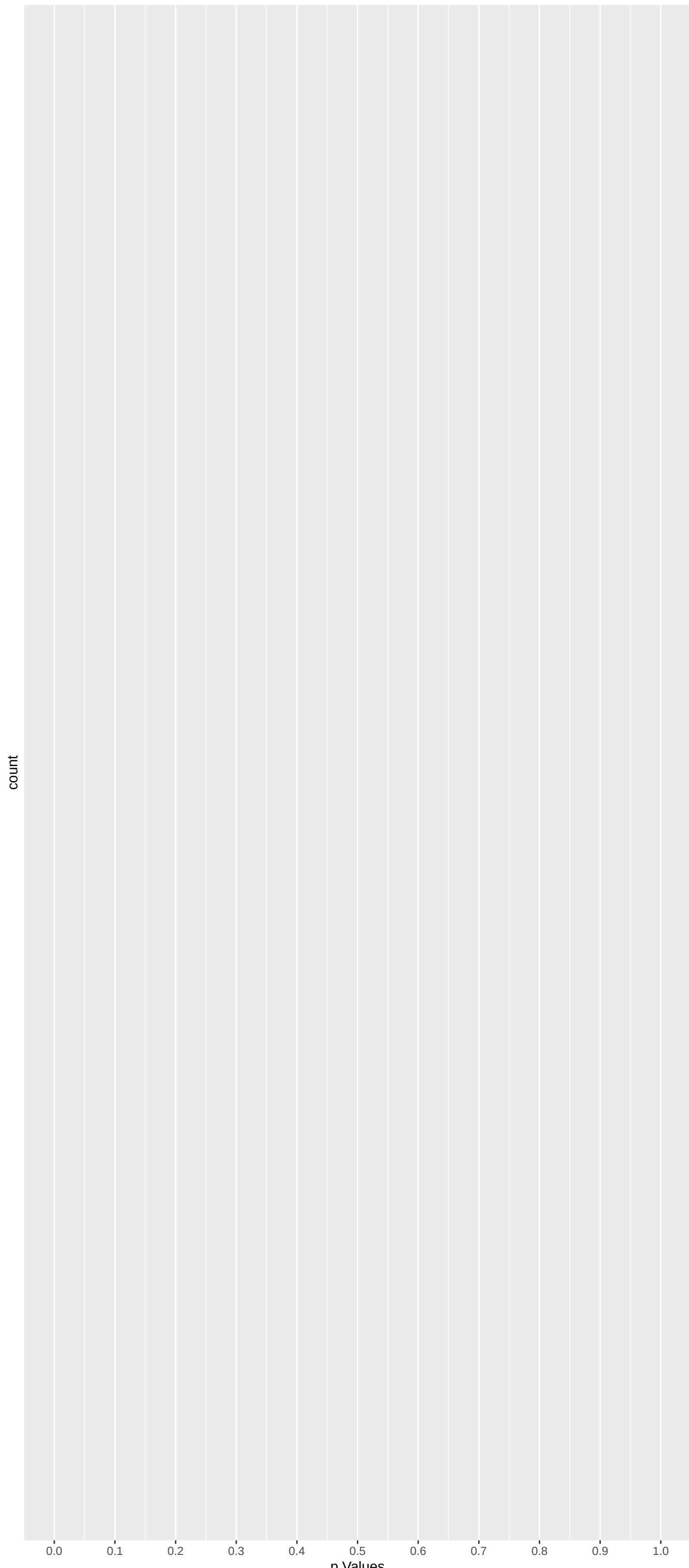
Positive Rho Non–permuted



Top Positive genes by P–value non–permuted

Gene	Rho	N	P	p.adj	qValueNoperm
SPG11	0.09921964	627	3.256e-14	5.252e-10	2.040e-06
TRPM5	0.11837287	511	7.777e-14	6.273e-10	2.040e-06
EP400	0.10360299	572	1.426e-13	7.669e-10	2.040e-06
ZFAND4	0.11180013	517	2.972e-13	1.199e-09	2.391e-06
TRIM15	0.13087578	423	1.016e-12	3.279e-09	3.432e-06
PLEC	0.11871026	461	1.388e-12	3.731e-09	3.432e-06
ADGRF1	0.08909133	604	2.160e-12	3.871e-09	3.432e-06
ECI2	0.10166850	530	2.095e-12	3.871e-09	3.432e-06
TMEM131L	0.09193039	590	1.746e-12	3.871e-09	3.432e-06
SLC14A2	0.09200406	573	3.735e-12	6.026e-09	4.421e-06

Negative Rho Non–permuted



Top Negative genes by P–value non–permuted

Gene	Rho	N	P	p.adj	qValueNoperm
NA	NA	NA	NA	NA	NA
NA.1	NA	NA	NA	NA	NA
NA.2	NA	NA	NA	NA	NA
NA.3	NA	NA	NA	NA	NA
NA.4	NA	NA	NA	NA	NA
NA.5	NA	NA	NA	NA	NA
NA.6	NA	NA	NA	NA	NA
NA.7	NA	NA	NA	NA	NA
NA.8	NA	NA	NA	NA	NA
NA.9	NA	NA	NA	NA	NA

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Gene Expression (GO:0010467)	-0.10052218	264	2.167e-08	1.168e-04	NLRP3:2 DEDD2:35 MRPS18C:71 NCBP:106 RPS13:196 MRPL34:202
mRNA Processing (GO:0006397)	-0.11420035	190	6.250e-08	1.684e-04	SF3B3:15 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292 LSM4:360
RNA Splicing, Via Transesterification Re	-0.11903688	160	2.189e-07	9.932e-04	SF3B3:15 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292 LSM4:360
mRNA Splicing, Via Spliceosome (GO:00003	-0.10285538	191	1.026e-06	1.382e-03	SF3B3:15 GEMIN4:101 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292
Peptide Biosynthetic Process (GO:0043043)	-0.11685336	134	3.152e-06	3.397e-03	MRPS18C:71 RPS13:196 MRPL34:202 MRPL12:235 MRPL10:312 RPLP0:328
Macromolecule Biosynthetic Process (GO:0	-0.10402129	159	6.388e-06	5.692e-03	MRPS18C:71 RPS13:196 MRPL34:202 MRPL12:235 LARGE1:276 MRPL10:312
Positive Regulation Of Transcription By	-0.04467808	879	9.623e-06	7.407e-03	NLRP3:2 TXK:3 BCL2L12:32 TRIAP1:33 HAX1:39 HES1:41
RNA Processing (GO:0006396)	-0.09617770	166	2.008e-05	1.082e-02	SF3B3:15 DEDD2:35 NCBP2:106 LSM1:113 EXOSC10:214 NUFP1:230
Inorganic Cation Transmembrane Transport	0.07494351	277	1.930e-05	1.082e-02	TRPM5:2 KCNK15:45 KCNK18:52 TRPM6:103 KCNV2:120 SLC6A6:125
Intermediate Filament Organization (GO:0	0.17474263	50	1.940e-05	1.082e-02	KRT80:228 KRT73:304 KRT32:326 BFSPP2:616 KRT13:1035 PKP1:1076
Positive Regulation Of Cell Motility And Tra	-0.03758071	1160	2.393e-05	1.172e-02	NLRP3:2 TXK:3 SF3B3:15 BCL2L12:32 TRIAP1:33 HAX1:39
Monatomic Cation Transmembrane Transport	0.07354281	273	3.137e-05	1.408e-02	TRPM5:2 KCNK15:45 KCNK18:52 TRPM6:103 KCNV2:120 SLC6A6:125
Vesicle Budding From Membrane (GO:0006090	-0.21775713	30	3.677e-05	1.524e-02	SNX3:168 TMED10:183 CHMP7:327 MYO18A:353 S100A10:489 FNBP1:1201
RNA Splicing (GO:0008380)	-0.12260892	86	8.667e-05	3.336e-02	SF3B3:15 NCBP2:106 LSM1:113 MAGOHB:282 LSM4:360 KHSPR:407
B Cell Homeostasis (GO:0001782)	-0.36612130	9	1.428e-04	5.128e-02	GAPT:45 TNFRSF13B:623 TNFAIP3:812 MEF2C:1250 PPP2R3C:1530 LYN:1816
Amino Acid Transport (GO:0006865)	0.16687718	43	1.543e-04	5.181e-02	SLC43A1:96 SLC6A6:125 SLC6A20:640 SLC6A15:867 SLC36A2:1318 SLC38A1:1446
Translation (GO:0006412)	-0.07644985	206	1.635e-04	5.181e-02	MRPS18C:71 RPS13:196 MRPL34:202 MRPL12:235 MRPL10:312 RPLP0:328
rRNA Metabolic Process (GO:0016072)	-0.12000399	80	2.104e-04	6.297e-02	DEDD2:35 EXOSC10:214 EBNA1BP2:224 NVL:303 EXOSC6:351 RPL27:496
Metal Ion Transport (GO:0030001)	0.08458162	157	2.641e-04	6.776e-02	CLDN16:21 KCNJ16:41 SLC17A4:50 KCNK18:52 SLC17A3:111 ATP2A1:278
Protein Localization To Centrosome (GO:0	0.25636597	17	2.532e-04	6.776e-02	CEP192:131 GOLGB1:434 KIAA0753:633 STL1:690 C2CD3:765 NUDCD3:1054
Regulation Of Small GTPase Mediated Sign	0.10124245	110	2.502e-04	6.776e-02	FGD5:22 TIAM2:82 ARHGEF19:110 SYDE12:180 FAM13B:371 RASGRF1:600
Regulation Of Transcription By RNA Polym	-0.02704343	1713	3.134e-04	7.675e-02	NLRP3:2 TXK:3 SF3B3:15 BRMS1L:17 BCL2L12:32 TRIAP1:33
Monocarboxylic Acid Transport (GO:001571	0.12884845	63	4.097e-04	9.598e-02	SLC27A4:151 ATP8B1:334 SLC16A5:337 SLC5A12:340 TSP02:531 SLC10A2:768
Potassium Ion Transmembrane Transport (G	0.08794635	132	4.882e-04	1.119e-01	KCNJ16:41 KCNK15:45 KCNK18:52 KCNV2:120 LRRC55:307 KCNF1:580
piRNA Processing (GO:0034587)	0.23332274	18	6.113e-04	1.318e-01	MOV10L1:135 PIWIL1:349 GATC2:562 ANKRD34B:553 PNDC1:1064 EXD1:1322
Lysosome Biogenesis (GO:0032418)	-0.16105531	37	7.027e-04	1.402e-01	TMEM106B:307 FCER1A:745 KIF5B:1195 PIK3CG:1455 TFEF1490 NDEL1:1850
Ribosome Biogenesis (GO:0042518)	-0.08353516	139	6.922e-04	1.402e-01	EIF2S1:47 RPS13:196 EXOSC10:214 EBNA1BP2:224 RPS19BP1:302 NVL:303
Negative Regulation Of Cell Motility And	-0.08677675	125	8.258e-04	1.589e-01	SF3B3:15 BRMS1L:17 TP53NP1:421 AMH:453 SLIT2:589 OGT:790
DNA Replication Checkpoint Signaling (GO	0.245758074	15	9.852e-04	1.659e-01	CLSPN:198 CDC45:381 ORC1:383 DNA2:1172 CDC6:1822 TIMELESS:2283
Negative Regulation Of Cell Migration (GO	-0.07307347	154	9.572e-04	1.659e-01	SF3B3:15 BRMS1L:17 TP53NP1:421 AMH:453 SLIT2:589 CLASP2:707
Sodium Ion Transport (GO:0006814)	0.10391530	85	9.415e-04	1.659e-01	SLC17A4:50 WKNA1:102 SLC17A3:111 SLC6A6:125 SCN10A:289 SLC5A12:340
Vasodilation (GO:0042311)	0.24583491	15	9.803e-04	1.659e-01	ADRB1:740 SOD1:1026 NOS3:1523 NOS1:2871 AGTR2:3209 SOD2:3219
Regulation Of Cell Proliferation And Prolif	-0.03663781	696	1.136e-03	1.855e-01	MTS1R:1 IFI251:29 HES1:41 PCOD10:48 IL31RA:78 IGFBP6:98
Cytoplasmic Translation (GO:0002181)	-0.10832109	73	1.391e-03	2.204e-01	RPS13:196 RPLP0:328 RPS5:452 RPL27:496 RPL6:593 RPL35A:714
Protein Peptidyl-Prolyl Isomerization (G	-0.24543711	14	1.476e-03	2.272e-01	PP1F:863 PP1G:1167 PP1L3:1240 PPLD:1959 PP1C:2243 PIN1:2444
Regulation Of mRNA Splicing, Via Spliceo	-0.10073673	83	1.532e-03	2.293e-01	CELF6:30 SRRM4:375 HNRNP1:475 HNRNPAL1:579 ZBTB7A:697 PTBP1:730
Monocarboxylic Acid Catabolic Process (G	0.20331876	20	1.649e-03	2.339e-01	LPIN1:226 CYP2BCL:268 ACOT8:609 HOGA1:712 LPIN2:875 AGXT:1969
Positive Regulation Of Interferon-Beta P	-0.16080771	32	1.649e-03	2.339e-01	IRF3:228 TLR4:416 RIKT3:704 PTPN11:763 IRF5:864 HSP90AA1:962
Regulation Of DNA-templated Transcriptio	-0.02391865	1646	1.723e-03	2.381e-01	SF3B3:15 DEDD2:35 HES1:41 CDK14:44 IRF2:50 YEATS4:61
Positive Regulation Of Transcription Ini	-0.12754104	50	1.822e-03	2.455e-01	NKX2-5:221 TAF10:429 MED7:1021 MED11:1071 MED16:1137 MED29:1148

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HOUNKPE_HOUSEKEEPING_GENES	-0.073436046	1061	1.198e-15	7.773e-12	STK38:8 CCT2:23 DERL1:24 CMPK1:34 HAX1:39 EIF2S1:47
REACTOME_METABOLISM_OF_RNA	-0.08325670	641	8.579e-13	2.783e-09	SF3B3:15 RAN:16 PSMD8:73 APOBEC1:85 GEMIN4:101 NCBP2:106
BLALOCK_ALZHEIMERS_DISEASE_DN	-0.06132198	1150	4.049e-12	8.756e-09	RSU1:6 LAPTM4B:9 CCT2:23 MAPK9:37 CDK14:44 EIF2S1:47
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	-0.07732462	692	5.598e-12	9.079e-09	NLRP3:2 CCS:27 DEDD2:35 MAPK9:37 EIF2S1:47 LMNB1:52
REACTOME_REGULATION_OF_MRNA_STABILITY_BY	-0.20730775	83	6.798e-11	8.821e-08	PSMD8:73 ELAVL1:142 PSMB5:223 ANP32A:261 HSPA8:292 PSMA1:344
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	-0.21648249	73	1.635e-10	1.769e-07	PSMD8:73 MAPRE1:134 PSMB5:223 TUBB2B:299 PSMA1:344 PSMC3:451
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	-0.05976798	1011	1.913e-10	1.773e-07	TRIAP1:33 HES1:41 YEATS4:61 GANT:64 CDC27:68 NBN:69
REACTOME_INFECTIOUS_DISEASE	-0.06601246	809	2.378e-10	1.929e-07	NLRP3:2 RAN:16 TAB2:19 PSMD8:73 GEMIN4:101 NCBP2:106
REACTOME_REGULATION_OF_EXPRESSION_OF_SL	-0.14962606	146	4.559e-10	3.286e-07	PSMD8:73 NCBP2:106 RPS13:196 PSMB5:223 ELOB:231 MAGOHB:282
KIM_BIPOLAR_DISORDER_OLGODENDROCYTE_DISE	-0.1076215	624	1.967e-09	1.276e-06	NSF:7 RAN:16 CLTB:18 CCT2:23 MAPK9:37 PPP6C:58
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	-0.12426449	189	4.051e-09	2.389e-06	PSMD8:73 NCBP2:106 RPS13:196 PSMB5:223 ELOB:231 MAGOHB:282
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	-0.04746257	1319	1.094e-08	5.913e-06	RSU1:6 LAPTM4B:9 SF3B3:15 CCT2:23 CMPK1:34 KCTD3:42
REACTOME_SIGNALING_BY_THE_B_CELL_RECEPTO	-0.15819368	105	2.189e-08	1.063e-05	PSMD8:73 BLK:147 ORAI1:217 PSMB5:223 PSMA1:344 SH3KBP1:422
PUJANA_BRCA1_PCC_NETWORK	-0.04406881	1479	2.294e-08	1.063e-05	ITPK1:11 RAN:16 CCT2:23 TRIAP1:33 CHST7:43 EIF2S1:47
MILI_PSEUDOPODIA_HAPTOTOXIS_UP	-0.07403883	485	2.690e-08	1.141e-05	PCDD10:48 MRPS18C:71 STYX:128 PIAR1:143 TIMM8A:195 SRP54:197
REACTOME_CLECTA_DECTIN_1_SIGNALING	-0.16582669	94	2.815e-08	1.141e-05	TAB2:19 PSMD8:73 PSMB5:223 CLEC7A:320 PSMA1:344 PSMC3:451
REACTOME_UCH7_PROTEINASES	-0.17278374	85	3.730e-08	1.407e-05	PSMD8:73 TPST1:129 PSMB5:223 PSMA1:344 PSMC3:451 USP15:517
BORCZUK_MALIGNANT_MESOTHELIOMA_UP	-0.09434338	288	3.905e-08	1.407e-05	RAN:16 CCT2:23 TRIAP1:33 MAPK9:37 CDC27:68 PSMD8:73
ACEVEDO_METHYLATED_IN_LIVER_CANCER_DN	0.06362738	637	5.063e-08	1.729e-05	FGD5:22 ADGRV1:31 ADGRG7:28 ANK1:42 TIAM2:82 SLC43A1:96
REACTOME_MRNA_SPLICING	-0.11347883	193	5.718e-08	1.855e-05	SF3B3:15 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292 LSM4:360
REACTOME_HIV_INFECTION	-0.10597977	214	9.619e-08	2.837e-05	RAN:16 PSMD8:73 NCBP2:106 PSMB5:223 ELOB:231 CDDB8:239
HSIAO_HOUSEKEEPING_GENES	-0.08545768	332	9.467e-08	2.837e-05	ITPK1:11 PSMD8:73 TMED10:183 PSMB5:223 COX41:287 RPLP0:328
REACTOME_FCR1I_MEDIATED_NF_KB_ACTIVATION	-0.17521528	77	1.079e-07	3.044e-05	TAB2:19 PSMD8:73 PSMB5:223 FCER1A:724 PSMA1:344 PSMC3:451
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	-0.16348035	88	1.174e-07	3.175e-05	PSMD8:73 PSMB5:223 PSMA1:344 CDK7:361 H4C1:393 PSMC3:451
PUJANA_CHEK2_PCC_NETWORK	-0.05850112	704	1.478e-07	3.836e-05	RAN:16 CCT2:23 TRIAP1:33 EIF2S1:47 PCDD10:48 LMNB1:52
REACTOME_SCF_BETA_TRCP_MEDIATED_DEGRADAT	-0.20898914	52	1.655e-07	4.003e-05	PSMD8:73 CDC20:111 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566
REACTOME_DNA_REPLICATION_PRE_INITIATION	-0.14200887	114	1.666e-07	4.003e-05	CDC27:68 PSMD8:73 PSMB5:223 PSMA1:344 H4C1:393 PSMC3:451
REACTOME_DOWNSTREAM_SIGNALING_EVENTS_OF_	-0.17345803	76	1.737e-07	4.025e-05	PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566 NFKBIA:799
WANG_TUMOR_INVASIVENESS_UP	-0.08111609	351	1.937e-07	4.201e-05	CCT2:23 STX7:72 NHP2:121 ELAVL1:142 SCAF8:145 COX41:287
REACTOME_C_TYPE_LECTIN_RECEPTORS_CLRS	-0.13942790	117	1.943e-07	4.201e-05	TAB2:19 PSMD8:73 PSMB5:223 FCER1G:274 CLEC7A:320 PSMA1:344
REACTOME_INNATE_IMMUNE_SYSTEM	-0.05235453	865	2.157e-07	4.488e-05	NLRP3:2 TXK:3 TAB2:19 CCT2:23 MAPK9:37 VPS35L:40
REACTOME_APC_C_MEDIATED_DEGRADATION_OF_C	-0.16458800	83	2.214e-07	4.488e-05	CDC27:68 PSMD8:73 CDC20:111 PSMB5:223 PSMA1:344 PSMC3:451
REACTOME_DECTIN_1_MEDIATED_NONCANONICAL_	-0.19534626	58	2.691e-07	5.146e-05	PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566 PSMA8:801
KEGG_SPliceOSOME	-0.14271748	109	2.697e-07	5.146e-05	SF3B3:15 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292 LSM4:360
REACTOME_TCR_SIGNALING	-0.14409078	106	3.031e-07	5.314e-05	TAB2:19 PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566
REACTOME_DISEASES_OF_CELLULAR_TRANSDUCTION	-0.07359314	415	2.986e-07	5.314e-05	HES1:41 PSMD8:73 NCBP2:106 CNKSR2:118 KREMLIN1:131 NR4A1:174
BENPORATH_MYC_MAX_TARGETS	-0.05756673	691	2.966e-07	5.314e-05	MTS1R:1 HAX1:39 EIF2S1:47 PCDD10:48 PPP6C:55 STX7:72
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	-0.10892158	186	3.133e-07	5.349e-05	PSMD8:73 BLK:147 PSMB5:223 PSMA1:344 CDK7:361 H4C1:393
REACTOME_MITOTIC_G1_PHASE_AND_G1_S_TRANS	-0.12342766	144	3.271e-07	5.442e-05	PSMD8:73 PSMB5:223 PSMA1:344 CDK7:361 PSMC3:451 MYBL2:553
MARTENS_RETINOIN_RESPONSE_DN	-0.05691750	699	3.475e-07	5.636e-05	RAN:16 TRIAP1:33 HES1:41 MRPS18C:71 PEX12:89 GEMIN4:101

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
22q11 Deletion Syndrome	0.28414095	19	1.810e-05	5.919e-02	HIRA:85 MJMD1C:603 CHR6:671 ZDHHC8:1370 DGCR8:1503 COMT:1536
Glioma	-0.03194195	1776	1.495e-05	5.919e-02	MST1R:1 RSU1:6 BCL2L12:32 CMPK1:34 MAPK9:37 HES1:41
Tetany	0.28378849	21	6.759e-06	5.919e-02	CLDN16:21 HIRA:85 TRPM6:103 MJMD1C:603 KCNJ1:644 SLC12A3:767
Glioblastoma	-0.03110326	1593	5.686e-05	1.116e-01	RSU1:6 BCL2L12:32 MAPK9:37 HES1:41 PCOD10:48 RARRES2:57
Precursor T-Cell Lymphoblastic Leukemia-Neoplasm Metastasis	-0.06466747	337	4.494e-05	1.116e-01	HES1:41 ZEB2:86 IGFBP7:119 TIMMBA:195 NKX2-5:221 CSA2:226
Anemia, hereditary spherocytic hemolytic	0.48861271	5	1.544e-04	1.515e-01	MST1R:1 RSU1:6 LAPTM4B:9 RAN:16 BRMS1L:17 COL6A3:21
Carcinogenesis	-0.02209973	3287	1.527e-04	1.515e-01	ANK1:42 SPTA1:70 SLC4A11:30 EPB42:275 SPTB:394 NA
Classical Hodgkin's Lymphoma	-0.09339394	142	1.263e-04	1.515e-01	MST1R:1 FSCN1:210 CYS1LFR1:238 TNFRSF17:315 EPB42:454 IL3RA:567
Spherocytosis	0.48861271	5	1.544e-04	1.515e-01	ANK1:42 SPTA1:70 SLC4A11:30 EPB42:275 SPTB:394 NA
Cytochrome-c Oxidase Deficiency	-0.17963093	36	1.929e-04	1.582e-01	CCS:27 COAT:136 COAE:159 COX41:287 ETHE1:592 COX5A:973
Lymphoma	-0.03432349	1072	1.936e-04	1.582e-01	MST1R:1 NLRP3:2 NSF:7 STK38:8 NBN:69 TNFSF13B:139
Malignant neoplasm of stomach	-0.02652887	1876	2.342e-04	1.767e-01	MST1R:1 STK38:8 LAPTM4B:9 COL6A3:21 CDK1:144 EIF2S1:47
Central neuroblastoma	-0.02918871	1407	3.443e-04	2.252e-01	NSF:7 RAN:16 MAPK9:37 HES1:41 IRF2:50 IL31RA:78
Squamous cell carcinoma	-0.02843021	1500	3.369e-04	2.252e-01	MST1R:1 MAPK9:37 HAX1:39 HES1:41 CDK14:44 EIF2S1:47
HIV Infections	-0.04054014	634	5.652e-04	3.465e-01	RAN:16 CMPK1:34 MAPK9:37 NBN:69 PSMD8:73 CCL19:179
22q11 partial monosomy syndrome	0.29166078	11	8.100e-04	3.484e-01	HIRA:85 MJMD1C:603 COMT:1536 ARVCF:1658 PI4KA:2297 TBX1:2862
Bilateral cataracts (disorder)	0.09504775	103	8.745e-04	3.484e-01	GSTO1:117 CRYGA:638 SMN2:499 BFSPP2:616 NFE2L2:664 SMN1:676
Broad chest	-0.35930702	7	9.945e-04	3.484e-01	BRAF:526 MAP2K1:675 PTPN11:763 CHST3:2186 DYM:3635 TRAPPCC2:3870
Familial intrahepatic cholestasis of pre	0.39443020	6	8.200e-04	3.484e-01	ATP8B1:334 ABCB8A:1161 NR1H4:1321 VCAM1:1624 ABCB11:1746 TNF:3765
Hamantoma of tongue	0.42522236	5	9.885e-04	3.484e-01	OFD1:692 C2CD3:765 WDPCP:1022 CTCN3:1434 NKX1:291
Leukemogenesis	-0.09994559	617	7.974e-04	3.484e-01	HAX1:39 HES1:41 NR4A1:174 NKX2-5:221 CDDB8:239 PRKAA2:255
Metabolic Bone Disorder	0.17754689	30	7.664e-04	3.484e-01	CA2:459 CYP2R1:482 PTH:1027 ESR1:1070 IL6:1085 MMP2:1182
Neuroblastoma	-0.02659929	1445	9.663e-04	3.484e-01	NSF:7 RAN:16 CMPK1:34 MAPK9:37 HES1:41 IRF2: